DC POWER SUPPLY

USER IS MANUAL

(SPS-W series)

VER:01

PRODUCT MAINTANCE

1. Disconnect the power when the product is not in use.
2. Unplug the power supply before cleaning.
3. Do not use hydrocarbons, chlorides or similar solvents, or use abrasive cleaners.

PRODUCT WARRANTY

1. This product is offered free maintenance service within one year from the date of purchase, Except in the following cases:
A: Lack of this product warranty card
B: Failures caused by improper use, such as improper handling and improper repair, modification or adjustment of the device.
C: Consumable materials are not covered by the warranty.
D: Naturally irresistible disasters such as floods, fires, earthquakes, etc.

2. Maintenance costs are charged for repairs that exceed the warranty period, and the costs incurred for maintenance are the responsibility of the user.

PACKING LIST

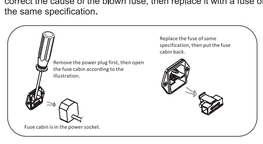
1. 1x Power Supply
2. 1x Power Cord
3. 1x Output Load Cord
4. 1x User's Manual
5. 1x Warranty Card

automatically switch to constant current mode. The output current remains stable and the output voltage decreases proportionally as the load increases further. The conversion of constant voltage and constant current is indicated by the LED on the front panel.

CV indicator light is on during constant voltage, CC indicator is on when constant current.

FUSE REPLACEMENT

If the fuse blows, the power supply will stop working. To find and correct the cause of the blown fuse, then replace it with a fuse of the same specification.



HIGH VOLTAGE! DANGER!

For effective safety protection, it is only necessary to replace the fuse of a specific specification. Before replacing the fuse, the power must be turned off and the power cord must be unplugged from the power outlet.

CONTENTS

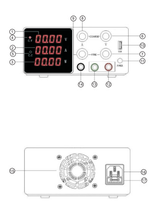
SAFETY BRIEF	1
SAFETY SYMBOL	1
PRODUCT BRIEF	2
PANEL INSTRUCTION	4
WORK REQUIEMENT	4
OPERATION INSTRUCTION	5
CONNECT THE LOAD	6
CONSTANT VOLTAGE/CONSTANT CURRENT CHARACTERISTICS	6
FUSE REPLACEMENT	7
PRODUCT MAINTENANCE	8
PRODUCT WARRANTY	8
PACKING LIST	8

Switchable DC regulated power supply

Model Number	SPS-W305/305D	SPS-W30/10/30/10D	SPS-W60/60/60D	SPS-W120/120D
Output Voltage	0~30V	0~30V	0~60V	0~120V
Output Current	0~5A	0~10V	0~5A	0~3A
Input Voltage：	230V±10% 50Hz（115V±10% 60Hz）			
Working Temperature：	0℃~40℃； Relative Humidity：＜80%RH			
Storage Temperature：	-10℃~70℃； Relative Humidity：＜70%RH			
Constant Voltage State：	Voltage stability≤0.1%+3mV Low Voltage:0.2~0.3%+3mV			
	Load stability≤0.5%+3mV			
	Ripple noises≤30mVrms			
Constant Current State：	Current stability≤0.2%+3mA			
	Load stability≤0.2%+3mA			
	Ripple noises≤20mArms（valid value）			
Display Accuracy：	0.5%+2digits			
Display Resolution：	Voltage:0.01V Current:0.001A			
Product Dimension：	long252mm X wide170X high 84			
Product Weight	1.2Kg	1.2Kg	1.2Kg	1.2Kg
Fuse Standard	3A(AC 220V Input)/5A(AC 110V Input)			

The above parameters are measured at an ambient temperature of 25± 5°C, relative humidity: ＜80%RH, and preheated for 30 minutes. The actual parameters will vary slightly.

PANEL INSTRUCTION



1. Output Voltage Display
2. Output Current Display
3. Output Power Display
4. Voltage Regulator
5. Current Regulator
6. Voltage Coarse Tuner
7. Voltage Fine Tuner
8. Current Coarse Tuner
9. Current Fine Tuner
10. USB Charging Socket
11. Power Switch
12. Positive Polarity (red)
13. Grounding (green)
14. Negative Polarity (black)

15. Cooling Fan
16. Power Socket
17. Fuse Box

WORK REQUIEMENT

1. AC input: Please make sure the input voltage of this product
   230V±10% 50Hz
   115 V±10% 60Hz
2. Do not use in an environment where the ambient temperature exceeds 40 degrees Celsius. The cooling fan is located at the rear of the device and should have enough space for cooling.

Warning: Incorrect AC voltage input will cause serious damage to the device. Please make sure the required input voltage value.

OPERATION INSTRUCTION

There are two types of power output modes: constant voltage output (CV) and constant current output (CC). The output mode is determined by the voltage and current values set by the user and the load connected by the user. The output voltage or current value of the power supply won't exceed the voltage and current values set by the user. In constant voltage mode, the output voltage value is equal to the user-set voltage value. In constant current mode, the output current value is equal to the user-set current value.

For example: the voltage value is set to 5V and the current value is set to 5A.
Steps
1. Turn on the powe switch
2. Adjust the voltage adjustment knob to 5V
3. Connect the "positive" and "negative" with wires and adjust the current knob to 5A.
4. Disconnect the wire and connect the load to use.

Attention: In actual CV operation, if the load resistance decreases and the output current increases to the set current value, the power supply will automatically switch to CC mode. When the load resistance value continues to decrease, the current will remain at the current set value. The voltage is proportionally reduced. At this time, increase the load resistance or increase the current set value to restore the CV output state.

PRODUCT BRIEF

The SPS-W series of adjustable regulated DC power supply designed for use in laboratories, schools and production lines. Both output voltage and output current are continuously adjustable between 0 and nominal.

The stability and ripple factor of the power supply are very good and have a perfect protection circuit. Can work at full load for a long time. This power supply can be used as both a regulated power supply and a regulated current supply.

SAFETY BRIEF

This manual contains important safety instructions that must be followed in the operation and storage environment of the SPS-W series. To ensure your personal safety, and ensures that this product works in the best environment, please read this manual carefully before using .

When you get a brand-new power supply, you need to do the necessary checks to make sure the instrument is working properly.
1. To check whether there are damages caused during transportation.
2. To check whether all the accessories are complete.
3. To check whether the output voltage and output current are normal after turning on the device.

If finding out any problems, please contact the merchant immediately.

SAFETY SYMBOL

The safety symbols below will appear in this manual or on the DC power supply.

Attention

High Voltage

Grounding

CONNECT THE LOAD

1. Rotate the terminal knob by turning it counterclockwise
2. Insert the load terminal
3. Turn the terminal knob clockwise
4. Banana plug can be directly inserted into the terminal hole



Attention: Improper connection may result in damage to the power supply and the load connected to the power supply. When connecting the battery load, do not reverse the polarity of the "+" and "-" as this may damage the power supply.

CONSTANT VOLTAGE / CONSTANT CURRENT CHARACTERISTICS

The working characteristics of this series of power supplies are constant voltage/constant current automatic conversion type, which can automatically change between constant voltage and constant current states with load changes. The intersection between constant voltage and constant current mode is called conversion point.For example, if the load causes the power supply to operate in a constant voltage mode, a constant voltage is output. As the load increases, the output voltage will remain constant and the output current will increase. When the current value reaches the set current limit value, the power supply will